

Constellation Highlight

CEPHEUS

The King

Husband to Cassiopeia, father of Andromeda, their story is a favorite legend among the constellations. It's the most northern constellation, aside from Ursa Minor, and has been in the list of constellations for millennia, included in Ptolemy's *Almagest*.

Before the conclusive reckoning of the "official" list of constellations, Cepheus shared a border with several now-ex-constellations:

Gloria Frederici - "Frederic's Glory" added by Bode (1781) for King Frederic of Prussia, carved out of the NW part of Andromeda

Custom Messium - the "Harvest Keeper" taken from Camelopardalis by Lalande (1775), to honor Charles Messier

Rangifer (also Tarandus) - the "Reindeer", also taken from Camelopardalis by LeMonier (1736) to commemorate an expedition by Maupertuis to Lapland (in arctic Scandinavia) - which given its northern declination seems appropriate on both counts.

Of course, none of these three constellations made the final list: still it's fun to try and locate their (always) dim stars, among the more-familiar constellations.

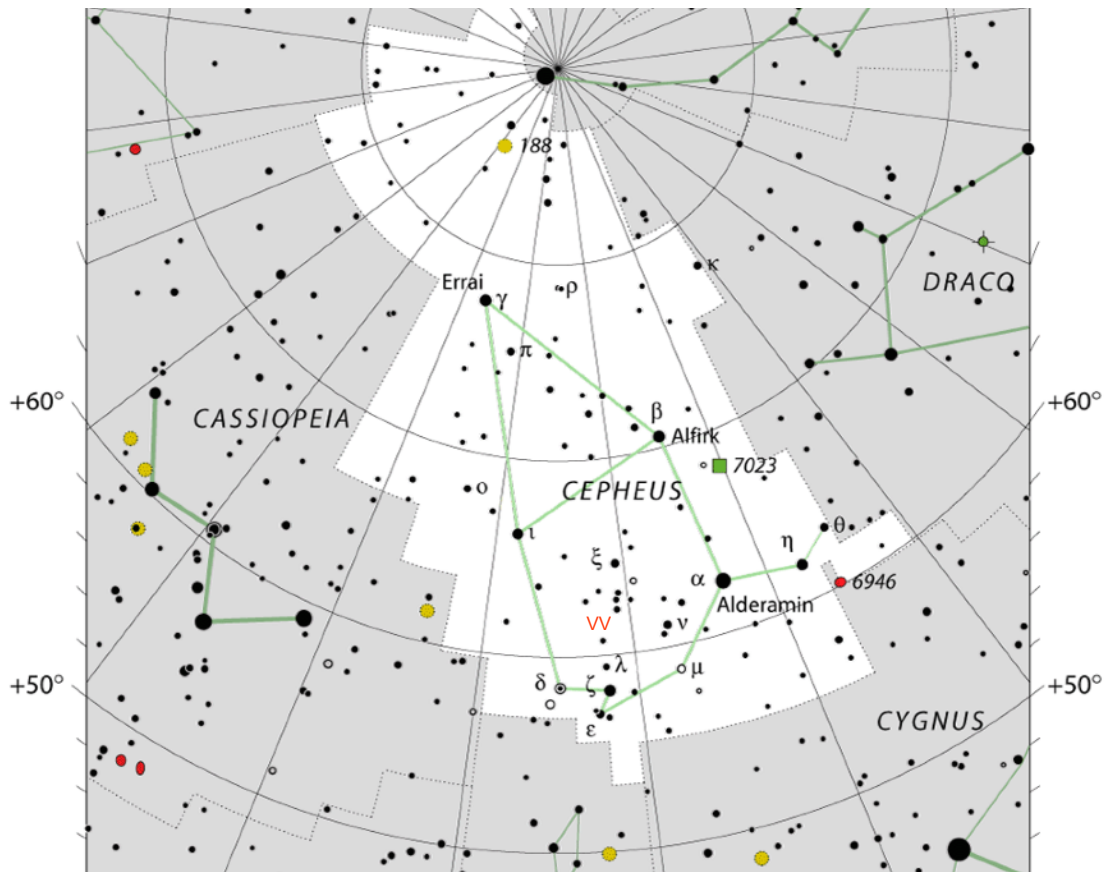
Cepheus's brighter stars form the stick-figure most resembling a doghouse, though in the depiction above, he is rendered upside-down, with the tip of the "house", γ (Gamma) Cephei as the king's left knee. Other atlases have him in a more regal pose with this star as his head.



The Northern Berkshire Astronomical Society

Meets the first Wednesday of every month, at the North Adams Public Library at 6 PM.

74 Church St. North Adams MA

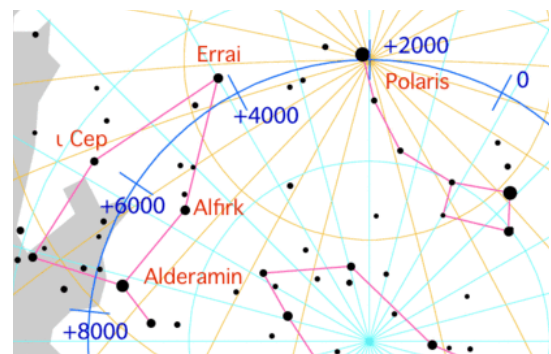


Map of Cepheus

The Milky Way crosses the south part of the constellation extending from Cygnus over to Cassiopeia: thus there are many open clusters, and nebulae of all types: emission nebulae, reflection nebulae, and dark nebulae, and a few planetary nebulae.

Alderamin (α , Alpha Cep) is the brightest star (mag 2.44). Mu (μ) Cephei is Herschel's "Garnet Star" and so large, in our Solar System would extend out beyond Jupiter! It's not the only noticeably "ruddy" star in the constellation: VV Cephei is another red supergiant (660-1000 times the size of the Sun) that eclipses its hot blue companion every 20.3 years (which is the 3rd longest eclipsing binary period known exceeded only by Epsilon Aurigae (27.08 yr) and AS Leonis Minoris (69.1 yr).

The identification of the "North Star" will include several stars in Cepheus over the next several millennia. The North Celestial Pole slowly moves away from Polaris, passing by Errai (Gamma Cep) 2000 years from now, then between Alfirk (Beta Cep) and Iota Cep, ~2000 years after that, then finally passing Alderamin (Alpha Cep) in another ~2000 years.

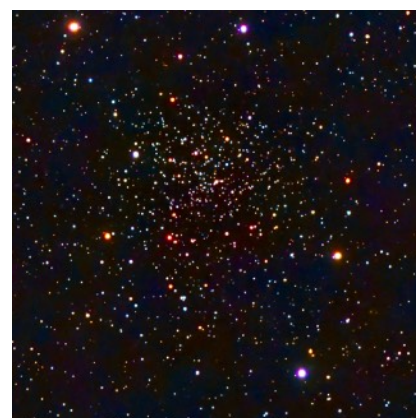


Things to See in Cepheus

Name	Type	Using
Caldwell 1 = NGC 188	Open Cluster	Medium/Imaging Telescope
Caldwell 4 = NGC 7023	Reflection + Dark Nebula	Binoculars/Small Telescope
IC 1396	Emission + Dark Nebulae	Small/Medium Telescope
Sh 2-136	Reflection Nebula	Imaging Telescope
NGC 7510	Open Cluster	Binoculars/Small Telescope
δ (Delta) Cephei	Variable Star	Naked Eye/Binoculars

Very Old Star Cluster

The open cluster closest to the celestial North Pole, which is why it was the first object in the Caldwell catalog (which runs north to south). However, this faint-but-rich cluster is more important because it is one of the very few older clusters - even older than the Sun (6.8 Gyr). It likely survived because of its location - it's far above the Galactic Disk, and further from the Galaxy's center than is the Sun.



Iris Nebula

The massive binary HD 200775 (mag 7.4), 1,350 ly away illuminates the surrounding dust grains to produce one of the most beautiful reflection nebulae in the sky. Aptly named the "Iris" Nebula (though it also looks like a pansy flower), the dust cloud which is 6 ly across shows a complicated structure. In small telescopes, the faint dark nebulosity looks more like an annulus of "missing" stars surrounding the nebula.

Elephant's Trunk

Almost 3,000 ly distant, this very complex nebulosity (this image only has the central part) is about 15-20 ly across. One curious feature (at about 1 o'clock in the image) is the "Elephant's Trunk" a dark globule (that might coalesce and form a new star).





Ghosts of Cepheus

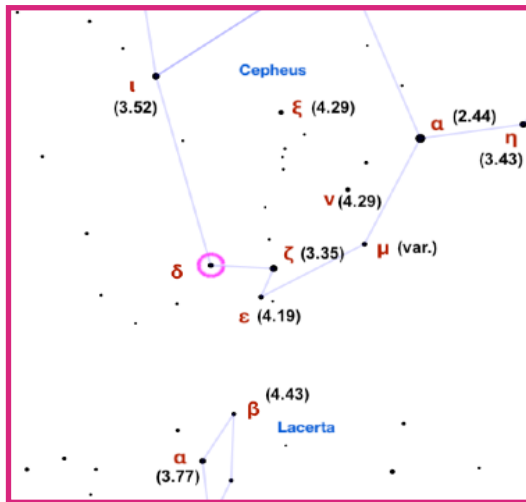
This eerie nebulosity is not only ghost-like, it almost looks organic in long exposures. The sprig-like features on the W side (upper left in this image) are reminiscent of the “Pillars of Creation” in the Eagle Nebula. At the opposite end, a denser globule appears to be pierced and split by an encroaching star. Definitely worth the time it takes to bring out all of the smoky detail.

Dormouse Cluster

This irregularly-shaped star cluster — like most open clusters — is young: only 10 Myr. However, it is somewhat distant: about 11.4 kyr, such that interstellar dust and gas dims the light of its stars by almost a magnitude. It’s almost known as the “Arrowhead” Cluster.

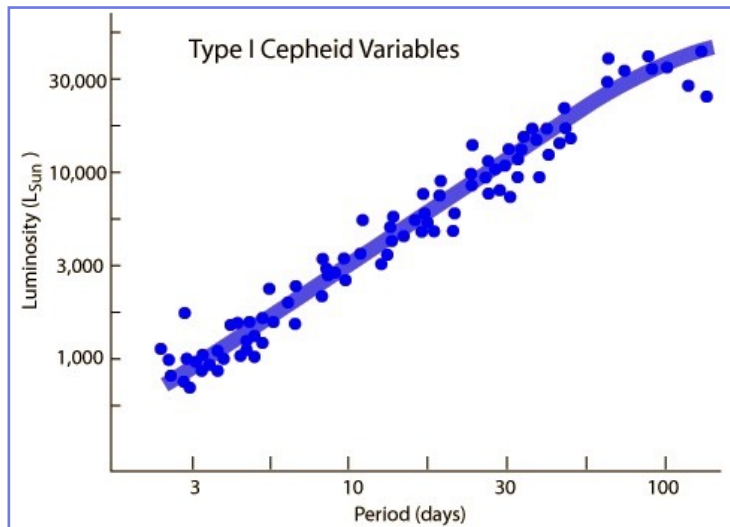
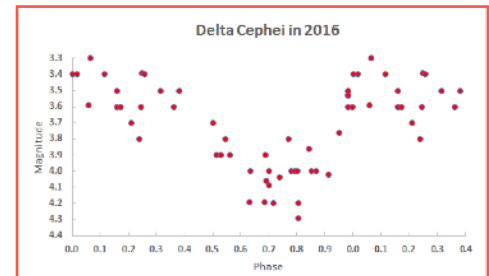


Important Variable Star



Cepheus’s 6th brightest star, δ (Delta, despite being the fourth Greek letter) is a variable star whose changes in brightness you can follow with the naked eye (or binoculars). It pulsates over a period of 5.37 days varying in brightness from mag 3.5 to 4.4.

Using this chart, you can compare the brightness of δ Cep to nearby stars in Cepheus and Lacerta.



Why Cepheid variables are important is their relationship between their pulsation period and their intrinsic brightness. When Hubble first identified a Cepheid variable in the Andromeda Galaxy, he was able to prove that galaxies were indeed “island universes” (of which the Milky Way was one) instead of “spiral nebulae” located within the Milky Way - one of the most important discoveries in the 20th century!